

DRAGONFALL

An Interactive Text-Based RPG



Submitted by:

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Submitted to:

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DRAGONFALL (Proposal)

A C++ console-based text-based Role-Playing Game (RPG). With its interactive level design, you can move in between dungeon rooms, fight bosses, talk to non-playable characters (NPC's) and finally slay the dragon. This game also features an intuitive easy to learn fight controls.

Main Features:

1. Immersive Story:

The game features a backstory of the main character, the world and the objective of the game making it more immersive and fun to play.

2. Interactive Level Design:

The gameplay is set in a dungeon which is made more interactive by giving the player choices on where to go next.

3. Attack System:

This game will feature an attack system used to fight against enemies and bosses.

4. NPC's:

Users can talk to NPC's which will give more context to the story and may give items to progress the story further.

5. Traps:

The Dungeon will have traps set up to make the gameplay more thrilling.

6. Special Items:

The player must gather a specific amount of a specific item to finish the game.

7. Boss Fight:

As the name suggests the player will have to slay the dragon to end the game.

8. Simple Console Menu:

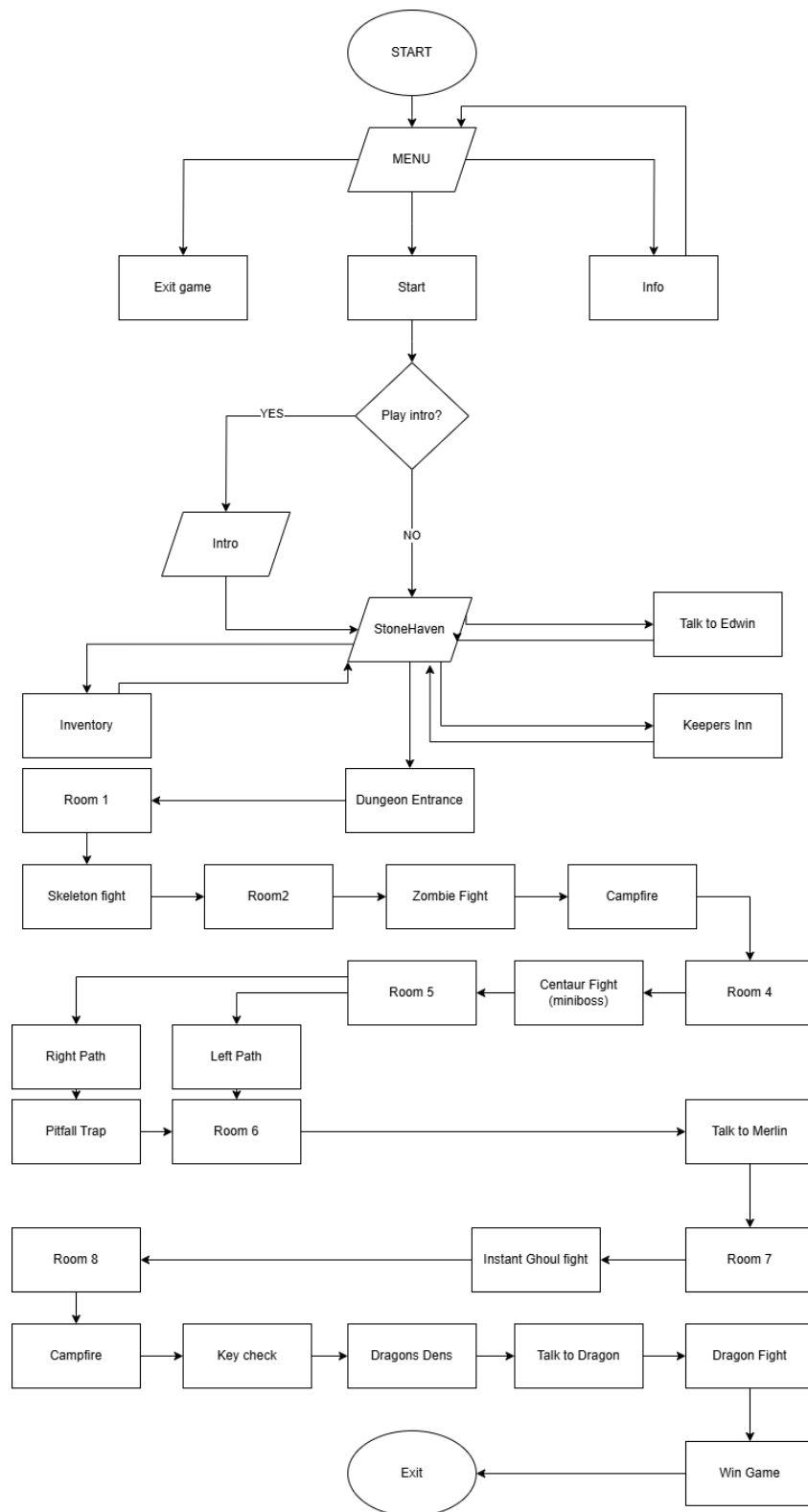
Boasting a simple, easy to understand console menu with different choices.

Project Members:

Muhammad Usman 24-SE-54

(Solo Project)

Flowchart:



Apart from the things mentioned in flowchart you can access inventory at almost every point checking Health, Attack, Defense, Level and Keys and make decisions in every room.

Game design:

Title Screen:

--DRAGONFALL--

- 1.Start
- 2.Info
- 3.Exit

Intro:

The game is set in a fantasy world called Aurendor where the humans and the undead are at war. The Undead while suffering defeat start sacrificing humans to give birth to The Great Dragon which is Evil Incarnate in itself. You play as a knight, the prophesized “golden child” who will defeat the dragon.

StoneHaven:

The Starting city where the Hero arrives. The city is located right outside of the dragons dungeon.

- Talk to Edwin.
- Keepers Inn.
- Enter Dungeon.
- Check Inventory.

Dungeon:

The Dungeon has an entrance, 8 rooms and a boss room.

Entrance:

- Final Warning to go into the Dungeon.

Room 1:

- Check inventory.
- Skeleton Fight.
- Level Up

Room 2:

- Light a Lamp.
- See a zombie in the light.
- Zombie Fight.
- Level Up

Room 3:

- Campfire (heal).
- Check Inventory.
- Warning to go into next room (miniboss – centaur).

Room 4:

- Centaur fight.
- Level Up.
- Get Green Key.
- Check Inventory.

Room 5:

- Choose a path.
- Left Path goes to next room.
- Right path goes to pitfall trap, then goes to next room.
- Check Inventory.

Room 6:

- Talk to Merlin.
- Get Enchanted by Merlin.
- Merlin gives you the Red Key.
- Level Up

Room 7:

- Automatic ghoul fight.
- Level up.
- Get Grey Key.

Room 8:

- Campfire (heal).
- Key Check.
- Warning to go to Dragons Den.

Dragons Den:

- Talk to the Dragon.
- Dragon Fight.

- Win the game.

Characters:

Grindell:

The Great human mage who went to war with the dragon many years ago. He was the reason humanity persists as he made a deal with the dragon, promising him a worthy challenger in return of him staying in the dungeon.

Edwin:

The owner of The Keepers Inn at StoneHaven. He oversees helping adventurers get ready to go into the dungeon.

Dragon:

The Great Dragon born from human sacrifices is evil incarnate itself and yearns a worthy challenger. He made a deal with Grindell that in a few years a prophesized hero would come and fight the Dragon.

HP 300

Attack 20

Defense 20

Merlin:

An Old Adventurer who lost his party and now sits in room 6 helping adventurers. Gives red key.

Mobs:

Skeleton:

First mob at the dungeon.

HP 30

Attack 5

Defense 5

Zombie:

Second enemy

HP 40

Attack 6

Defense 6

Centaur:

Miniboss (gives green key)

HP 100

Attack 10

Defense 15

Ghoul:

Miniboss (gives grey key)

HP 150

Attack 15

Defense 15

CODING TECHNIQUES:

Functions:

Used functions to make code less redundant.

- Talk to Edwin
- Talk to Merlin
- Inn (for healing)
- Fight
- Inventory

Choices:

Using the if and if else statements to make choices.

Arrays:

Arrays are used to store player and enemy information to make less variables.

e.g. `skeleton[3]={30,5,5};`

Random Number Generator:

Uses a Random Number Generator to either:

- Execute Attack.
- Critical attack.
- Attack miss.

Clear Screen:

Clearing screen using:

```
cout << "\033[2J\033[H";
```

033 is an ANSI escape code, [2J selects screen and clears it, [H at current position.

Time Dilation:

Using:

```
this_thread::sleep_for(chrono::seconds(1));
```

we can add pauses of set number of seconds in the code.

Header Files:

Used the following Header Files:

`<iostream>` for basic I/O.

<iomanip>

<thread>To use Time dilation.

<chrono>For Time dilation and srand().

<cstdlib>For rand().

Same Variable:

Used same Variable as a selection variable for all choices.

Global Variables:

The following are global variables for easy use:

- Selection.
- Enemies
- RNG.